

Analog Spiking Neuron in 28 nm CMOS

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Abstract—Traditional computer clusters are facing a significant limitation as a result of the big data revolution. We need efficient edge devices to bring the power of machine learning algorithms from power-hungry room servers to mobile consumer platforms. Neuromorphic engineering is a promising avenue for developing the next generation of edge devices that combine high computing capabilities with low power consumption in a small form factor. This paper shows the proof of concept of an analog/mixed-signal CMOS neuromorphic system on a chip (NeuroSoC) by presenting a low-power design of a leaky integrate-and-fire (LIF) neuron. The design uses eight transistors and two capacitors for low complexity and potential to lead to very dense systems. The proposed model consumes 1.2 fJ/spike and occupies an active area of 6.73 μm by 5.09 μm when implemented in 28 nm CMOS. The maximum spiking frequency is 343 kHz.

Index Terms—Neuromorphic, analog CMOS, leaky-integrate-and-fire (LIF) model, machine learning, spiking neural networks (SNN)

I. INTRODUCTION

Modern graphics processing unit (GPU) architectures have contributed significantly to the current artificial intelligence (AI) research renaissance throughout the past few decades. GPUs are now more accessible to the general public as a result of their low cost. The recent revolution in deep learning is redefining the nature of computing, which is driven by a surge in pattern classification, cognitive tasks, and massive datasets. These days, machine learning models are frequently deployed in the cloud, where they are trained and fine-tuned.

However, real-time applications are run on the edge and executed on mobile consumer platforms. Wireless communications, such as 5G, bridge the powerful remote ML models and the target application gap.

Nevertheless, the demand for more specialized computing platforms is unavoidable due to the ever-increasing trend in data and models. The goal is to transfer the machine learning model to a mobile platform and perform local prediction computations. In this regard, neuromorphic computing is a viable option for developing faster and more energy-efficient edge computing devices.

Figure 1 depicts a design flow that explains the methods utilized in neuromorphic engineering. By definition, we exploit an efficient natural process by abstracting the concept into a mathematical model and implementing it at the system level.

As the fundamental component of a neural network, the neuron cell is vital to overall performance. An efficient way

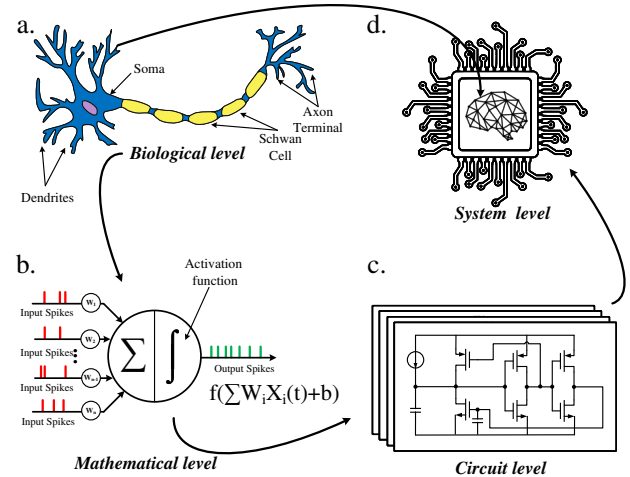


Fig. 1: Design flow of bio-inspired artificial neurons: a) Simplified diagram of a biological neural cell; b) abstract model with spiking output; c) circuit implementation; d) system level implementation for real-life application.

to formalize the neuron's dynamics is the leaky integrate-and-fire (LIF) model [1], which closely mimics the behavior of a biological neuron. A neuron is stimulated or inhibited by the outputs of other neurons through synapses, where neurotransmitter chemical compounds mediate the accumulation of charges in the cell's body via ionic channels. When the accumulation reaches a threshold, the ionic channels are activated to form an action potential or spike at the neuron's axon, and the neuron returns to its resting state. This process is modeled in the LIF by a leaky integrator coupled with a trigger mechanism to elicit the typical response; the resultant spike then propagates to other LIF neurons.

The LIF model is seen in conventional ANNs as a suitable compromise between the complexity of biologically plausible neuron models such as the Hodgkin-Huxley and Izhikevich models [2] [3]. The LIF model retains the analog nature of biologically plausible neuron models, thus allowing an efficient design of neuromorphic hardware. It also avoids the Von Neumann bottleneck paradigm, in which memory and processing units are still separated, while being simple to train with appropriate backpropagation techniques [4]. Neuromorphic hardware has led to the development of new materials for the fabrication of electrical components that mimic the

capacity of synapses to hold varying resistor values, hence modifying the signal accordingly. To create reliable energy-efficient neuromorphic systems in CMOS technology [5], such components still face challenges that must be addressed. Using smaller transistors to decrease operational voltage, hence reducing power consumption, is another method. It significantly minimizes the total area occupied by electronic components, making it the de facto solution for deploying real-time applications on edge devices. However, the reduction in voltage also diminishes the trans-impedance gain, requiring the insertion of buffer circuits for correction, and nullifies the size gain. Therefore, caution is required when employing this method.

In this paper, the design of an analog spiking neuron, based on the LIF model, using TSMC 28 nm CMOS technology is presented. Section II explains the role of each design block and the construction of the test benches used to validate the proposed circuit. The post-layout simulations are presented in Section III, followed by a discussion and comparison with recently reported designs. The paper concludes with Section IV, where we resume the highlight of the work presented throughout the paper.

II. MATERIALS AND METHODS

A. Materials

Many works have been published over the years to model the spiking behavior of biological neurons. The most complex one is the Hodgkin-Huxley (HH) model, which consists of a set of nonlinear differential equations that simulate the electrical properties of excitable cells like neurons. Izhikevich proposed a intermediate model that accounts for various spiking and bursting behaviors with only two differential equations [3]. This model has already been implemented as a CMOS circuit [6], but no large-scale system-on-chip has been developed using this model yet. On the other end of the spectrum, the LIF is the simplest model to emulate a biological neuron, as it only accounts for the leaky integration and the spiking threshold. The simplicity of this model makes it promising for the implementation of large-scale NeuroSoC processors in real-life applications, as is the case for the IMB TrueNorth chip [7].

Figure 2 depicts the suggested neuron design. Three blocks constitute the circuit. The first block (in orange) implements a basic current mirror. According to Equation (2), the transistor M_2 replicates the current I_{syn} flowing through the drain of transistor M_1 , which yields the current in the drain of M_2 :

$$I_{D2} = \frac{1}{2} K_p \left[\frac{W_2}{L_2} \right] [V_{GS} - V_{th}]^2 [1 + \lambda V_{DS}] \quad (1)$$

where K_p is a technology-related constant associated with the transistor, $\frac{W_2}{L_2}$ is the width-to-length ratio of transistor M_2 , V_{GS} is the gate source voltage, V_{th} is the threshold voltage, λ is the channel length modulation constant, and V_{DS} is the drain-source voltage.

The drain of the transistor M_1 models the synaptic gap between two neurons and where neurotransmitters are flowing.

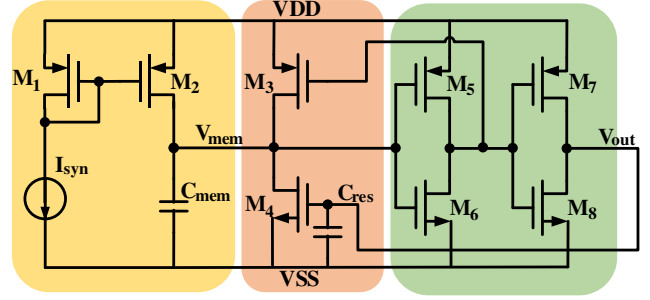


Fig. 2: Neuron circuit architecture. Orange: Integration block; Red: Reset block; Green: Spike generation block.

The capacitor C_{mem} integrates the input synaptic current which progressively increases the membrane voltage V_{mem} . The potential at this node increases linearly when there is a constant non-zero input current following equation (3). The period T is variable and depends on the input flow. Since the data flow in SNNs is intrinsically intermittent in space and time, we may never accurately estimate the value of T . Instead, the integration period is limited by the triggers set in the following blocks.

$$I_{syn} = C_{mem} \frac{dV_{mem}}{dt} \quad (2)$$

$$V_{mem} = \frac{1}{C_{mem}} \int_T I_{syn} dt \quad (3)$$

The third block (green) is made up of two back-to-back inverters that are used to generate spikes or artificial action potentials at the output of the neuron. Each generated spike encodes all integrated input spikes through the membrane capacitance C_{mem} . Although this processing seems simple, the reality is not so trivial, and there is a second temporal dimension to consider. In fact, the inter-periods between spikes at the input are processed through the neuron to generate output spikes at given times and intervals.

To understand the process of generating spikes by the third block highlighted in green, the evolution of the state variable V_{mem} must be monitored. When V_{mem} increases to a preset threshold level V_{Spike} , the inverters are engaged, resulting in a single spike at the output. The value of the spiking threshold voltage V_{Spike} is determined by the switching voltage of the first inverter composed of M_5 and M_6 in equation (4). In this case, the two inverters operate as a voltage driven one-shot circuit such that :

$$V_{Spike} = \frac{[V_{thn} + \frac{V_{Dsatn}}{2}] + r [VDD + V_{thp} + \frac{V_{Dsatp}}{2}]}{1 + r} \quad (4)$$

where V_{thn} and V_{thp} are threshold voltages, V_{Dsat} is the saturation voltage of the drain source and r is a technology variable where $r = \frac{K_p V_{DSATn}}{K_n V_{DSATp}}$ [8].

The second block highlighted in red, composed of transistors M_3 , M_4 and C_{res} , is activated when a spike is generated at the output. This reset circuit acts as a discharging mechanism to bring the membrane voltage V_{mem} to its resting potential, which is about 10 mV to 20 mV. The accuracy of the resting potential is not critical to the proper operation of the spiking neuron as much as the offset voltage in a typical operational amplifier. When V_{mem} returns to its resting potential, the cycle of integration, comparison, and spike generation can start over.

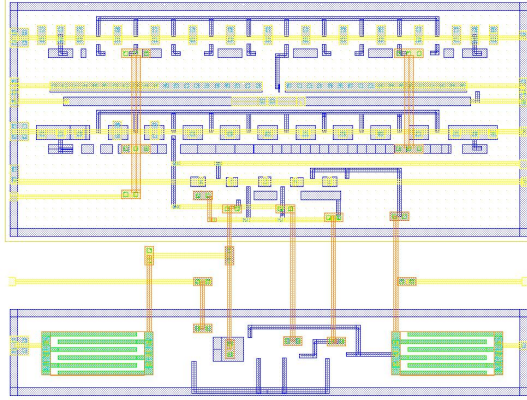


Fig. 3: Footprint of the neuron circuit implemented using the Cadence Virtuoso Layout Suite. the layout area is rounded $5 \mu m$ by $6 \mu m$.

B. Methods

To characterize the circuit model, several tests were performed to verify the consistency of the circuit implementation with the biological behaviors required to design large-scale NeuroSoCs. Simulations were performed at the post-layout stage after extraction of parasitic capacitors and resistors. The extraction was performed after the addition of dummy metal fills for more accurate simulation results.

1) **Spiking behavior over time:** In a biological neuron, the shape of the spike is caused by the instant bursting flow of ions across the axon membrane. The circuit implementation proposed in this paper shows a similar behavior when the synaptic input current integrated in C_{mem} reaches a certain threshold.

To carry out this experiment, we stimulated the neuron at its input with an ideal current source. Figure 4 depicts the excitatory stimulation. The amplitude of the current is $300 pA$. An intermittent stimulus signal is used to ensure that the neuron only fires when an excitatory current is applied and that no spikes are generated otherwise. To test this hypothesis, the output node of the proposed neuron is probed, and the resultant artificial action potentials are analyzed.

2) **Evolution of membrane potential:** By integrating the input synaptic current through the capacitor, the suggested neuron circuit in this paper exhibits the same behavior as a biological neuron. C_{mem} models the evolution of the membrane voltage V_{mem} . To perform this experiment, the neuron is excited with the same input stimulus as in the previous experiment and the internal membrane voltage is probed at the positive end of the membrane capacitor C_{mem} .

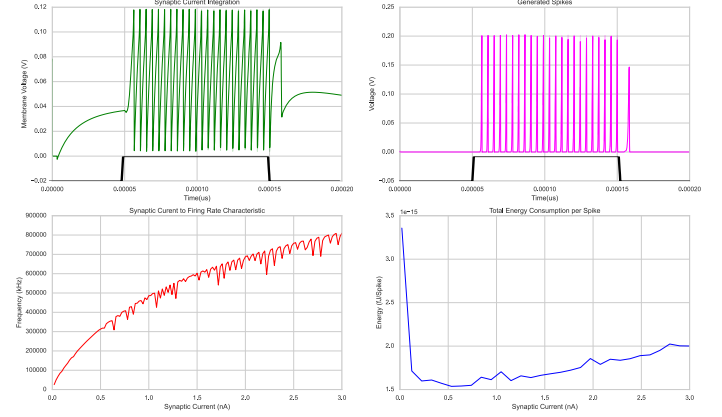


Fig. 4: Transient simulation of the membrane voltage in green for an input current of $300 pA$. Transient simulation of the output voltage in magenta: Regular spiking of the neuron is obtained for an excitatory current of $300 pA$. Characteristic simulation of the spiking frequency in red and energy consumption per spike over the synaptic current

3) **Energy consumption per event:** NeuroSoCs are coveted for their low power and high density. The proposed LIF neuron implementation's energy consumption is analyzed to evaluate this factor. The current drawn by the circuit during operation is measured, and the power supplied over the time the neuron is spiking is calculated. In parallel, the mean spike frequency throughout the same time period is computed. Finally, the power consumption is divided by the frequency of spiking to produce the energy consumption per spike of the proposed architecture. By varying the input synaptic current, the same experiment is repeated in a systematic manner. The characteristic of the energy consumption drawn by the neuron per generated spike vs. the input synaptic current is obtained by assessing this factor over a dynamic range of excitation current. The synaptic current dynamic ranges from $10 pA$ to $3000 pA$.

4) **Spiking behaviour versus input synaptic current:** Another interesting feature for validating the proposed design is to repeat the initial spiking experiment by increasing the input current and examining the mean spiking frequency at each step. The generated waveform shall reflect the frequency of the neuron's spiking behavior over the synaptic current dynamic range. The design's overall layout footprint in Figure 3 is rounded to $5 \mu m$ by $6 \mu m$. Layout techniques such as dummy transistors and active region balancing have been used to compensate for process variation. In sub-40nm CMOS technologies, special effects should not be neglected. By using large transistors at the input, the mismatch between the two branches of the input current mirror block is reduced. Given that the information is stored in the temporal coding between spikes rather than the precise value of synaptic current amplitude integrated in C_{mem} , the mismatch becomes an offset that can be deduced and corrected via subsequent calibration.

Parameters	This design*	Indiveri et al. 2006 [9]	Joubert et al. 2012 [10]	Cruz-Albrecht et al. 2012 [11]	Wijekoon et al. 2008 [12]	Danneville et al. 2019 [13]	Sourikopoulos et al. 2017 [14]	Ferreira 2019 [15]	Ronchini et al. 2020 [6]
Technology	28 nm	0.35 μm	65 nm	90 nm	0.35 μm	65 nm	65 nm	55 nm	180 nm
Supp. Volt. (V)	0.2	3.3 V	1 V	1.2 V	3.3	0.2	0.2	(-0.1,0.1)	0.2
Implementation	Analog	Analog	Analog	Analog	Analog	Analog	Analog	Analog	Analog
Neuron type	LIF	LIF	LIF	LIF	Izhikevich	A-H	ML-based	ML-based	Izhikevich
Time scale	Accelerated	Biological	Accelerated	Biological	Accelerated	Biological	Biological	Accelerated	Biological
Mem. Cap. (F)	3.4 fF	432 fF	500 fF	-	100 fF	5 fF	4 fF	9*934 aF	35 fF
Energy (J/spike)	1.2 fJ	900 pJ	2 pJ	400 fJ	9 pJ	2 fJ	4 fJ	3.6 fJ	58.5 fJ
Spiking frequency	343 kHz	200 Hz	1.9 MHz	100 Hz	1 MHz	15.6 kHz	26 kHz	360 kHz	-
Area/neuron (μm^2)	34	2,573	120	442	2800	34	35	85	472

TABLE I: Comparison of different neuron implementations in CMOS technology. *Post-layout simulations

III. RESULTS AND DISCUSSION

The proposed spiking CMOS neuron model is implemented in TSMC 28 nm CMOS technology. The design has eight transistors and two capacitors, which minimizes the complexity of the design. The silicon neuron has a surface area of $34 \mu\text{m}^2$. The simulations were conducted after the layout phase including all additional extracted parasitic impedance. The results presented were obtained with a supply voltage of 200 mV.

1) **Spiking behaviour over time:** As illustrated in Figure 4 in magenta, artificial action potentials are generated at the output whenever a current is present at the input. The amplitude of the spike is around 170 mV and the missing 30 mV accounts for the saturation of the drain-to-source voltage.

2) **Evolution of the membrane potential:** The evolution of the membrane potential is illustrated in Figure 4 in green. In the absence of an excitatory current, the membrane voltage remains at a constant voltage level of 35 mV. The presence of a current induces the integration of the membrane voltage, and, once it reaches a predefined threshold, an output spike is generated. Both the membrane voltage and the output voltage are then reset to a resting potential.

3) **Energy consumption per event:** This neuron has an energy efficiency of 1.2 fJ/spike. This value is obtained for a synaptic current of 30 pA. The black curve in Fig. 4 illustrates the energy consumption per spike through the dynamic range of the spiking neuron.

4) **Spiking behaviour versus input synaptic current:** The evolution of the spiking frequency is highly correlated with the input current, as shown in Figure 4. The peak value reaches 334 kHz. This performance is achieved by using small capacitors. The characteristic curve presented in red in Figure 4 can be approximated with a square-root function plus constants.

Table I shows a comparison between the performance of the published spiking artificial neurons. We can notice that a shrink in technology may not mean a gain in performance such as in [13], [14], [6] and this design.

This design has a lower energy consumption metric than the state-of-the-art, but the active area footprint is the same. Furthermore, the maximum spiking frequency exceeds the current state-of-the-art. However, the timescale in this design is shorter than in [13], [14], [6]. In comparison to an accelerated temporal design, such as [15], where maximal spiking rates

are comparable regardless of membrane capacitance and active silicon footprint. Be that as it may, the methods used to compute energy consumption in [14] and this work are not similar.

IV. CONCLUSION

In this paper, a leaky integrate-and-fire neuron model is proposed in 28 nm CMOS technology. The novelty of the proposed circuit is the analog implementation of a spiking artificial neuron using conventional 28 nm CMOS technology. In addition, the energy consumption per event can reach $1.2 \text{ fJ}/\text{Spike}$ and the maximum frequency of spiking is 343 kHz. The simulation results are evaluated after the layout extraction step. The design yielded an active surface of $34 \mu\text{m}^2$. These results provide additional evidence that the proposed neuron is a promising architecture for large-scale NeuroSoCs.

The design's primary characteristics are ultra-low energy usage and a small active footprint. These properties make the neuron suitable for large-scale implementation of neuromorphic system-on-chip, in which weighted synapses and neurons interact to generate complex inference-based dynamics and bring the power of machine learning algorithms to the edge.

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